Appendix. Class-wise evaluation of lithology-identification performance for the common component of MWD signals (per-class metrics and confusion matrices)

Section 4 of the main text summarizes the performance of six classifiers on the original signal and on the common component in terms of overall accuracy. This appendix reports class-wise results: precision, recall and F1 for each lithology, macro-averaged F1, balanced accuracy, and the confusion matrices. The protocol is a class-stratified 80/20 split, grid search under 5-fold cross-validation on the training set, and the mean over 10 repeats, with the same random seeds, splitting routine and search grids as in the main text. Because the five lithologies are not equally represented, metrics that weight each class equally are reported here alongside overall accuracy.

1.1 Class distribution of the samples

Table 1. Sample distribution of the five lithologies for each feature combination.

Signal	State	Sandstone A	Sandstone B	Sandstone C	Granite	Limestone	Total
Pressure	Original signal	29	5	11	15	11	71
Pressure	Common component	833	26	105	200	110	1274
Torque	Original signal	31	6	11	16	11	75
Torque	Common component	915	28	127	240	110	1420
Pressure + torque	Original signal	34	6	11	15	16	82
Pressure + torque	Common component	832	30	110	210	130	1312

The lithologies differ in the number of available boreholes, and pairwise pairing expands the sample size in proportion to roughly the square of that number, so lithologies with more boreholes expand faster and the class distribution of the common component is less even than that of the original signal.

1.2 Macro-averaged F1 and balanced accuracy

The two tables below are laid out as Table 9 of the main text, so that they can be compared cell by cell.

Table 2. Macro-averaged F1 of each classifier (mean ± standard deviation over 10 repeats, %).

Signal	State	LDA	KNN	SVM-RBF	XGBoost	LightGBM	Random Forest
Pressure	Original signal	65.0 ± 13.5	75.1 ± 10.2	83.0 ± 7.2	80.1 ± 12.0	78.3 ± 12.8	83.6 ± 10.9
Pressure	Common component	71.3 ± 3.7	87.4 ± 2.9	95.9 ± 1.8	95.9 ± 5.1	95.9 ± 3.3	96.5 ± 1.8
Torque	Original signal	52.5 ± 7.1	75.8 ± 13.1	65.2 ± 13.3	72.2 ± 10.4	66.6 ± 10.3	72.1 ± 8.5
Torque	Common	58.2 ± 3.3	81.9 ± 5.6	90.0 ± 3.8	93.8 ± 1.9	94.8 ± 1.1	94.8 ± 1.8

	component						
Pressure + torque	Original signal	55.9 ± 11.9	74.9 ± 11.4	76.5 ± 13.5	83.1 ± 15.8	76.0 ± 15.9	94.6 ± 5.3
Pressure + torque	Common component	76.7 ± 3.5	84.5 ± 3.5	93.7 ± 2.8	97.7 ± 0.8	97.6 ± 2.3	97.8 ± 1.3

Table 3. Balanced accuracy of each classifier (mean ± standard deviation over 10 repeats, %).

Signal	State	LDA	KNN	SVM-RBF	XGBoost	LightGBM	Random Forest
Pressure	Original signal	69.0 ± 12.9	78.3 ± 9.9	84.5 ± 6.5	83.1 ± 11.0	81.3 ± 11.9	84.9 ± 10.6
Pressure	Common component	68.8 ± 2.9	85.2 ± 3.3	95.1 ± 2.6	95.3 ± 5.7	94.8 ± 4.5	95.4 ± 2.6
Torque	Original signal	59.7 ± 8.6	81.0 ± 11.8	70.0 ± 12.5	76.0 ± 11.5	69.3 ± 10.9	77.0 ± 8.9
Torque	Common component	57.6 ± 3.3	80.4 ± 4.9	88.4 ± 4.0	92.5 ± 2.7	94.4 ± 1.8	93.4 ± 1.9
Pressure + torque	Original signal	62.3 ± 11.2	75.7 ± 11.5	78.0 ± 13.5	85.3 ± 13.4	78.7 ± 15.3	94.7 ± 5.2
Pressure + torque	Common component	77.1 ± 3.4	83.5 ± 4.4	92.1 ± 3.6	97.2 ± 1.3	96.9 ± 3.3	97.3 ± 1.7

Both metrics fall below the corresponding overall accuracy, as expected of equally weighted metrics under unequal class sizes: Sandstone B, the smallest class, carries the same weight as Sandstone A, the largest. The conclusion does not change with the metric. Across all 18 comparisons (3 feature combinations × 6 classifiers), the macro-averaged F1 of the common component exceeds that of the original signal without exception, by 3.3 to 28.2 percentage points. Balanced accuracy improves in 15 comparisons; the remaining three (pressure–LDA, torque–LDA and torque–KNN) show small decreases of 0.2 to 2.1 percentage points, all confined to linear discriminant analysis and nearest neighbours, the two models already weakest on the original signal, and the macro-averaged F1 of those three still improves, indicating that the slight loss of recall is offset by a gain in precision. The repeat-to-repeat standard deviation is also markedly smaller for the common component, that of macro-averaged F1 falling from 5.3–15.9 to 0.8–5.6 percentage points, so classification on common-component features is considerably less sensitive to the particular data split.

1.3 Class-wise metrics and confusion matrices

Table 4 gives the class-wise metrics of the best model for each state under the pressure + torque combination, and Figs. 1 to 3 give the confusion matrices for the pressure, torque and pressure + torque combinations in turn, each using the most accurate model for that combination.

Table 4. Class-wise precision, recall and F1 of the best model under the pressure + torque combination (pooled over 10 repeats, %).

State	Lithology	Support	Precision	Recall	F1
Original signal (Random Forest)	Sandstone A	60	95.2	98.3	96.7

	Sandstone B	10	100.0	100.0	100.0
	Sandstone C	20	100.0	95.0	97.4
	Granite	30	96.0	80.0	87.3
	Limestone	30	88.2	100.0	93.8
Common component (LightGBM)	Sandstone A	1660	99.5	99.8	99.6
	Sandstone B	60	100.0	90.0	94.7
	Sandstone C	220	99.1	98.6	98.9
	Granite	420	98.8	97.4	98.1
	Limestone	260	95.9	98.8	97.3

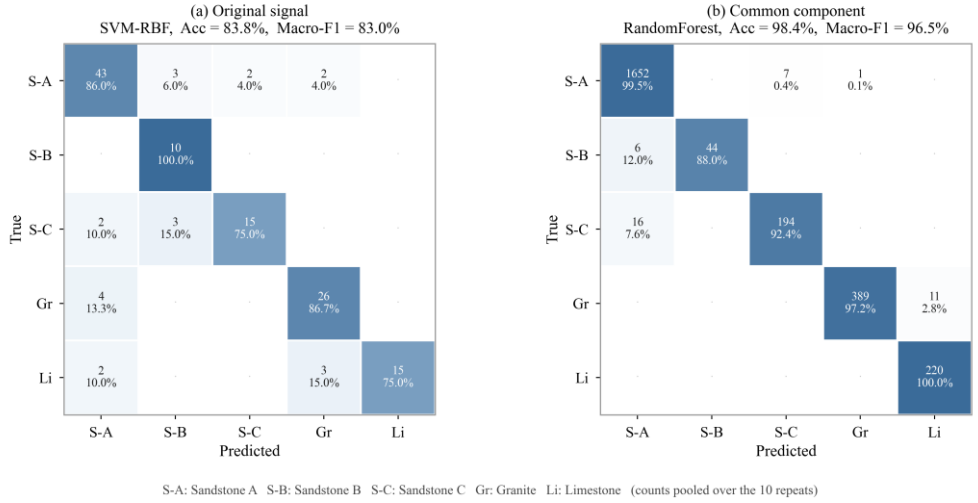


Fig. 1. Confusion matrices for the original signal and the common component under pressure features (pooled over 10 repeats; the best model for that combination is used in each case).

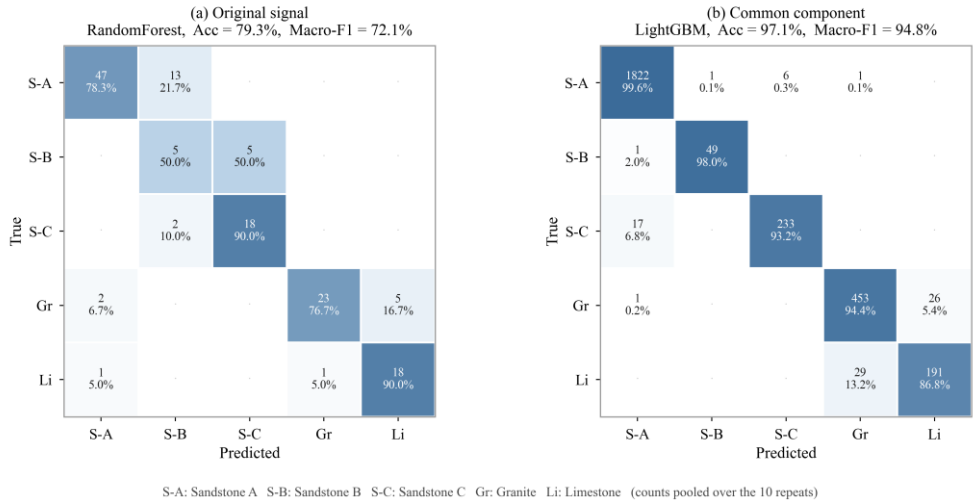


Fig. 2. Confusion matrices for the original signal and the common component under torque features (pooled over 10 repeats; the best model for that combination is used in each case).

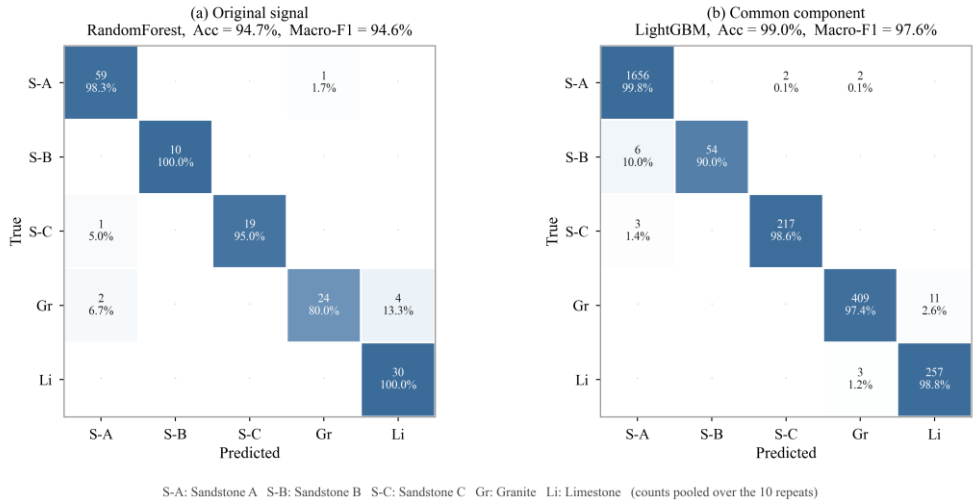


Fig. 3. Confusion matrices for the original signal and the common component under pressure + torque features (pooled over 10 repeats; the best model for that combination is used in each case).

Under the common component the F1 of all five lithologies is at least 94.7%. The lowest belongs to Sandstone B, the smallest class, with precision 100.0% and recall 90.0%, its misclassifications falling entirely into Sandstone A. Next is the mutual confusion between Granite and Limestone: the recall of Granite is 97.4%, with 11 samples assigned to Limestone. The latter agrees with the analysis in Section 5.1 of the main text, where granite and limestone are close in uniaxial compressive strength (81.90 MPa and 72.22 MPa) and their common-component amplitudes therefore differ little. On the original signal, by contrast, the recall of Granite is only 80.0%, the main source of its low macro-averaged F1, which the common component raises to 97.4%.

1.4 Conclusions

Under metrics that weight each class equally, the advantage of the common component over the original signal still holds and is in fact clearer. The macro-averaged F1 improves in all 18 comparisons without exception, by 3.3 to 28.2 percentage points; balanced accuracy improves in 15, and the small decreases in the remaining three all occur in the linear discriminant and nearest-neighbour models that were already weakest on the original signal, whose macro-averaged F1 still improves.

The common component also markedly reduces the sensitivity of the results to the data split, the repeat-to-repeat standard deviation of macro-averaged F1 falling from 5.3–15.9 to 0.8–5.6 percentage points, so its discriminative information does not rest on any particular partition of the samples.

The class-wise results show that under the common component the F1 of all five lithologies is at least 94.7%, with the residual error concentrated in two places: Sandstone B, the smallest class, whose misclassifications fall entirely into Sandstone A, another sandstone; and the mutual confusion between Granite and Limestone, which are close in strength. Both are explicable from the rock properties themselves rather than from any defect of the decomposition method, and the latter corroborates the discussion of common-component amplitude mapping in Section 5.1 of the main text.